Jin Schofield

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EDUCATION

Computer Science and Mathematics, Senior | Princeton University | Sept. 2022 - May 2026

- 1590 SAT (800 Math, 790 Reading/Writing - 99th percentile), 36 ACT (100th percentile), 44/45 IB Diploma (99th percentile).
- President of Princeton NeuroTech, Officer of Princeton AI, Principal at Princeton Student Ventures (VC Club), Pitched to GenNTech, AstraZeneca, Trinity Life Sciences with E-Club.
- Coursework: Advanced Algorithm Design (graduate course), Optimization, Abstract algebra, Distributed Systems, Probability Theory, Mathematical game theory, Non-Euclidean geometry, Reinforcement learning, Theory of algorithms, Operating Systems, Theory of natural algorithms (graduate course), Probability + stochastic systems, Real analysis, Functional programming, Data structures and algorithms, Introduction to programming systems, Introduction to machine learning, Reasoning about computation, Cognitive neuroscience, Linear algebra

SKILLS

- Skills: Jax, PyTorch, Optuna, CUDA, Numpy, MLFlow, AWS, Python, C++, C, OCaml, Java, R, Keras, NumPy, Pandas, Git, MediaPipe

EXPERIENCE

Machine Learning Research Engineer Intern | Allen Control Systems | Austin, Texas | June 2025 - Present

- Re-architected and trained large core ML object detection model, allowing for detection of drones in additional ~50% of camera FOV that was previously too noisy, for autonomous precision robotics system for drone interception using recently published research papers while decreasing latency by ~75% compared to research paper implementation. Constructed a pipeline for fine-grained control of hyper-realistic synthetic training data generation using recently published research papers, increasing accuracy by additional 5%.

Reinforcement Learning Researcher | Princeton Reinforcement Learning Lab and Griffiths Lab, Princeton University | May 2025 - Present

- Co-authored and engineered for BuilderBench paper. Website: https://rajghugare19.github.io/builderbench/. Arxiv: https://arxiv.org/abs/2510.06288.
- Created uniformity-only contrastive representation learner that rivals the accuracy of a variational auto-encoder while using 11% fewer parameters by eliminating the need for a decoder. Project progress visible here. Currently researching exploration for contrastive reinforcement learning using temporal representations.

Teaching Assistant for COS 226 and COS 217 | Princeton University CS Department | January 2025 - May 2025

- Taught students data structures and algorithms and programming systems.

Machine Learning Researcher (Latent Diffusion) | Ramaswamy ML Interpretability Lab, Princeton University | December 2024 - April 2025

- Developed novel conditional diffusion method for cellular automata synthesis in Conway's Game of Life. Project write-up here. GitHub https://github.com/jinschofield/Gen-GOL
- Researched application of diffusion on language with Diffusion of Thoughts

Quantitative Research and Data Science Intern | QuantCap, LLC | December 2024 - January 2025

- Used RNNs, XGBoost, Random Forest to generate strategy for 305% cumulative returns on options over 12-month period. Supervised and mentored by Quantitative Researcher at BlackRock, ex-Meta Data Scientist, UPenn Wharton. Constructing data sets with SQL and Python based on Black-Scholes model.

AI Research and Software Engineering Intern | CalTech Perona Computer Vision Lab | June 2024 - Aug. 2024

- Implemented novel in-context learning in latent diffusion models with few-shot method for unsupervised keypoint generation. Optimized detection of specific key points with fewest training data possible. Project write-up here. Improved accuracy of detection of specific key points by 26% using only 5 sample training images

Product Intern | GPTZero | June 2023 - September 2023

- I designed features for educational AI tools at GPTZero, an AI edtech start-up with $10 million in funding. Wrote a weekly newsletter for 45 000 subscribers.

Data Science/ML Intern | Oxford University Clinical Research Unit | July 2023 - August 2023

- Analyzed extensive patient datasets using R. Enhanced predictive accuracy of patient outcomes using ML.

Software Engineering Intern | AfimaCheck | June 2021 - August 2021

- Programmed back-end C++ code of SeekThermal infrared cameras for immunoassay bodily fluid THC detection device. Programmed GUI in Python.

PUBLICATIONS

BuilderBench — A Benchmark for Generalist Agents | 2025 https://rajghugare19.github.io/builderbench/ https://arxiv.org/html/2510.06288v1

- Co-authored benchmark and evaluated performance of various RL algorithms on general 3D block structure construction tasks.

PROJECTS NOT ASSOCIATED WITH AN INTERNSHIP

AIstotle - An LLM-powered educational chrome extension for YouTube videos | 2023

- Uses GPT-4 Vision, ElevenLabs, and retrieval-augmented generation.

ConchShell - A wearable ASL translator that uses AI to convert ASL letters into a voice | 2022

- Engineered a wearable device using C++ and PyTorch to translate ASL letters into speech, achieving 94% translation accuracy and securing $16600 in funding through DMZ accelerator support, while gaining national recognition.

AWARDS & HONORS

1st Place Jane Street Estimathon | Harvard Undergraduate Trading Competition

Represented Princeton at ICPC Regionals 2024, 2025 | ICPC Regionals

97th Percentile Fryer Math Contest | University of Waterloo